Comprobación ejercicio 5 laboratorio 4

Laboratorio 4 ejercicio 5

$f(x) = 2x^3 - 11.7x^2 + 17.7x - 5$

Método de punto fijo:

$f(x) = 2x^3 - 11.7x^2 + 17.7x - 5$

$x_0 = 3$

3 iteraciones

Transformando ecuación:

$2x^3 - 11.7x^2 + 17.7x - 5 = 0$

$x = \frac{5 + 11.7x^2 - 2x^3}{17.7}$

Iteración #1 $i = 0$

Paso 1: $x_{i+1} = g(x_i)$

$x_1 = \frac{5 + 11.7(3)^2 - 2(3)^3}{17.7}$

$x_1 = 3.181$

Paso 2: $\varepsilon_a = \left| \frac{x_{0+1} - x_0}{x_1} \right| 100\%$

$\varepsilon_a = \left| \frac{3.181 - 3}{3.181} \right| 100\%$

$\varepsilon_a = 5.69\%$

Iteración #2 $i=1$

Paso 1: $x_{1+1} = g(x_1)$

$x_2 = \frac{5 + 11.7(3.181)^2 - 2(3.181)^3}{17.7}$

$x_2 = 3.334$

Paso 2: $\varepsilon_a = \left| \frac{x_{1+1} - x_1}{x_2} \right| 100\%$

$\varepsilon_a = \left| \frac{3.334 - 3.181}{3.33} \right| 100\%$

$\varepsilon_a = 4.59\%$

Iteración #3 $i=2$

Paso 1: $x_{2+1} = g(x_2)$

$x_3 = \frac{5 + 11.7(3.334)^2 - 2(3.334)^3}{17.7}$

$x_3 = 3.443$

Paso 2: $\varepsilon_a = \left| \frac{x_{2+1} - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \frac{3.443 - 3.334}{3.443} \right| 100\%$

$\varepsilon_a = 3.17\%$

Paso 2: $\varepsilon_a = \left| \frac{x_{0+1} - x_0}{x_1} \right| 100\%$

$\varepsilon_a = \left| \frac{3.181 - 3}{3.181} \right| 100\%$

$\varepsilon_a = 5.69\%$

Iteración #2 $i=1$

Paso 1: $x_{1+1} = g(x_1)$

$x_2 = \frac{5 + 11.7(3.181)^2 - 2(3.181)^3}{17.7}$

$x_2 = 3.334$

Paso 2: $\varepsilon_a = \left| \frac{x_{1+1} - x_1}{x_2} \right| 100\%$

$\varepsilon_a = \left| \frac{3.334 - 3.181}{3.33} \right| 100\%$

$\varepsilon_a = 4.59\%$

Iteración #3 $i=2$

Paso 1: $x_{2+1} = g(x_2)$

$x_3 = \frac{5 + 11.7(3.334)^2 - 2(3.334)^3}{17.7}$

$x_3 = 3.443$

Paso 2: $\varepsilon_a = \left| \frac{x_{2+1} - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \frac{3.443 - 3.334}{3.443} \right| 100\%$

$\varepsilon_a = 3.17\%$

Método de Newton - Raphson:

$f(x) = 2x^3 - 11.7x^2 + 17.7x - 5$

$X_0 = 3$

$e = 0.001$

$f'(x) = 6x^2 - 23.4x + 17.7$

Iteración #1 $i = 0$

Paso 1: $f(x_0) = 2(3)^3 - 11.7(3)^2 + 17.7(3) - 5 = -3.2$

$f'(x_0) = 6(3)^2 - 23.4(3) + 17.7 = 1.5$

Paso 2: $X_{0+1} = X_0 - \frac{f(x_0)}{f'(x_0)}$

$X_1 = 3 - \frac{-3.2}{1.5}$

$X_1 = 5.133$

Paso 3: $\varepsilon_a = \left| \frac{X_{0+1} - X_0}{X_1} \right| 100\%$

$\varepsilon_a = \left| \frac{5.133 - 3}{5.133} \right| 100\%$

$\varepsilon_a = 41.55\%$

Iteración #2 $i = 1$

Paso 1: $f(x_1) = 2(5.133)^3 - 11.7(5.133)^2 + 17.7(5.133) - 5 = 48.07$

$f'(x_1) = 6(5.133)^2 - 23.4(5.133) + 17.7 = 55.67$

Paso 2: $X_{1+1} = X_1 - \frac{f(x_1)}{f'(x_1)}$

$X_2 = 5.133 - \frac{48.07}{55.67}$

$X_2 = 4.269$

Paso 3: $\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_2} \right| 100\%$

$\varepsilon_a = \left| \frac{4.269 - 5.133}{4.269} \right| 100\%$

$\varepsilon_a = 20.24\%$

Iteración #3 $i=2$

Paso 1: $f(x_2) = 2(4.269)^3 - 11.7(4.269)^2 + 17.7(4.269) - 5 = 12.94$

$f'(x_2) = 6(4.269)^2 - 23.4(4.269) + 17.7 = 27.15$

Paso 2: $x_{2+1} = x_2 - \frac{f(x_2)}{f'(x_2)}$

$x_3 = 4.269 - \frac{12.94}{27.15}$

$x_3 = 3.792$

Paso 3: $\varepsilon_a = \left| \frac{x_{2+1} - x_2}{x_3} \right| 100\%$

$\varepsilon_a = \left| \frac{3.792 - 4.269}{3.792} \right| 100\%$

$\varepsilon_a = 12.58\%$